Yueyao Yuan (袁月耀)

Incoming M.S. in Computer Science, Zhejiang University | Sept 2026

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Education

Zhejiang University — M.S. in Computer Science and Technology Sept 2026 – Jun 2029 (expected)

Advisor: Prof. Zhe Liu (post-quantum cryptography) • Co-advisor: Dr. Xiaogang Xu (Zhejiang Lab; multimodal LLM & alignment)

Shandong University — B.Eng. in Data Science and Big Data Technology (Qingdao) Sept 2022 – Jun 2026

GPA 4.17 / 5.0 (91.7 / 100), Rank 3 / 45 • Thesis: "Self-Rewriting Prompt Methods for LLMs" (advisor: Prof. Jianhua Yin)

Research Interests

Large language models: **post-training and reinforcement learning for reasoning** — self-distillation, credit assignment, training stability (entropy dynamics), retrieval-augmented reasoning, and alignment.

Publications & Preprints

Bold = me.

Under Review & In Preparation

[1] **Yueyao Yuan**, et al. *Epistemic Tokens as Functional Anchors: Why Self-Distillation Fails in Mathematical Reasoning and How to Fix It.* Under review, *EMNLP 2026*.

[2] Jiale Cui, **Yueyao Yuan**, Xiaokun Mu, Xiaogang Xu, Zhe Liu, Jiafei Wu, Lu Zhou. *PQC-LLM: A Hierarchically Adapted Post-Quantum Cryptography Model.* Manuscript, 2026.

[3] **Yueyao Yuan**, et al. *DivGRPO: Divergence-Guided Asymmetric Credit Assignment for Reasoning RL.* In preparation, *AAAI 2027*.

Research Experience

Masked Self-Distillation for Reflective Mathematical Reasoning — *lead* (EMNLP 2026 submission) 2025 – 2026

- Found that on-policy self-distillation **degrades reflective reasoning**: a reference-conditioned teacher systematically suppresses the student's **epistemic tokens** (e.g. "Wait") that express uncertainty and drive self-verification.
- Proposed **masked self-distillation** — a minimal change that excludes epistemic tokens from the distillation loss, preserving reflective behavior; consistently beats naive self-distillation on **AIME24, AIME25, MATH-500**.

DivGRPO: Divergence-Guided RL for LLM Reasoning — *lead* 2026 – present

- Diagnosed GRPO's uniform credit assignment (final-outcome reward broadcast to **all** tokens) as a driver of entropy instability on distilled reasoning models.
- Proposed **DivGRPO**: **asymmetric, multiplicative** reweighting of the advantage at token-level divergence points (pure token space; no extra model/hidden states). On **verl** across DeepSeek-R1-Distill 1.5B/7B and Qwen3-4B, it gives stable low-entropy convergence and preserves AIME accuracy where GRPO destabilizes.

Selected Projects

LLM from scratch: pre-training → SFT → RLHF

- Implemented Transformer, tokenizer, and the full pre-training / SFT / PPO-GRPO pipeline from scratch — hands-on intuition for reward design, KL control, and entropy monitoring.

Honors & Awards

- First-class Scholarship (×1) and Second-class Scholarship (×2), Shandong University
- Merit Student (三好学生), Shandong University
- Honorable Mention, Mathematical Contest in Modeling (MCM)
- CCF CSP: 250 (top 6%)
- Undergraduate major rank **3 / 45**; GPA 4.17 / 5.0 2022–2026

Technical Skills

- **Programming**: Python (proficient), C/C++, SQL
- **Machine Learning**: PyTorch, LLM pre-training / SFT / RLHF (PPO, GRPO), verl; RAG, information retrieval; distributed training (FSDP), CUDA
- **Tooling**: Linux, tmux, Git; vLLM / SGLang inference
- **Languages**: Chinese (native), English (CET-6 518; academic reading & writing)